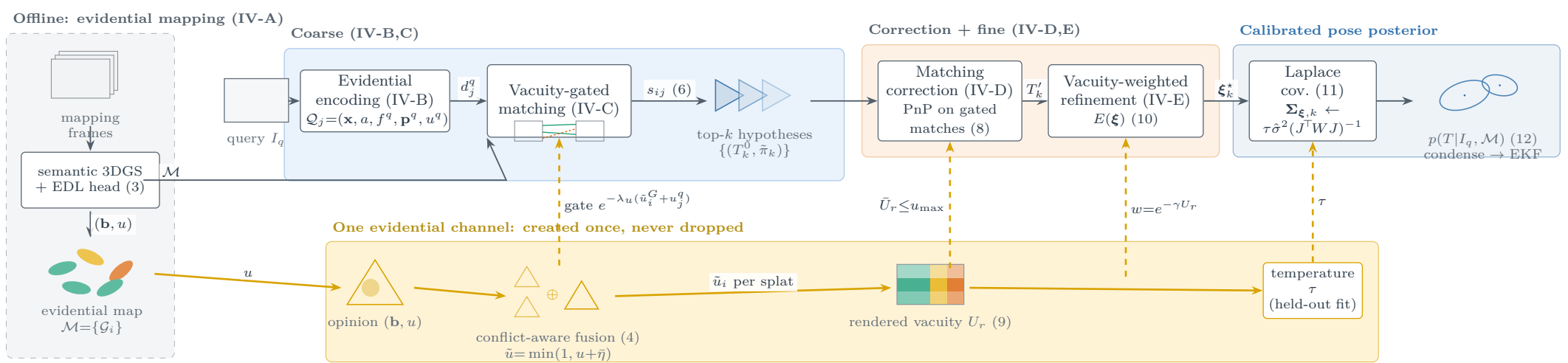
Offline: evidential mapping (IV-A)
mapping frames
semantic 3DGS + EDL head (3)
$\mathcal{M}$
$(\mathbf{b}, u)$
evidential map $\mathcal{M}=\{\mathcal{G}_i\}$
Coarse (IV-B,C)
query I_q
Evidential encoding (IV-B) $\mathcal{Q}_j=(\mathbf{x}, a, f^q, \mathbf{p}^q, u^q)$
d_j^q
Vacuity-gated matching (IV-C)
s_{ij} (6)
top-k hypotheses $\{(T_k^0, \tilde{\pi}_k)\}$
Correction + fine (IV-D,E)
Matching correction (IV-D) PnP on gated matches (8)
T_k'
Vacuity-weighted refinement (IV-E) $E(\boldsymbol{\xi})$ (10)
$\boldsymbol{\xi}_k^\star$
Calibrated pose posterior
Laplace cov. (11) $\boldsymbol{\Sigma}_{\boldsymbol{\xi},k} \leftarrow \tau\hat{\sigma}^2(J^\top W J)^{-1}$
$p(T|I_q, \mathcal{M})$ (12) condense → EKF
gate $e^{-\lambda_u(\bar{u}_i^G + u_j^q)}$
$\bar{U}_r \le u_{\max}$
$w = e^{-\gamma U_r}$
τ
One evidential channel: created once, never dropped
u
opinion $(\mathbf{b}, u)$
conflict-aware fusion (4) $\tilde{u} = \min(1, u + \bar{\eta})$
$\tilde{u}_i$ per splat
rendered vacuity U_r (9)
temperature τ (held-out fit)